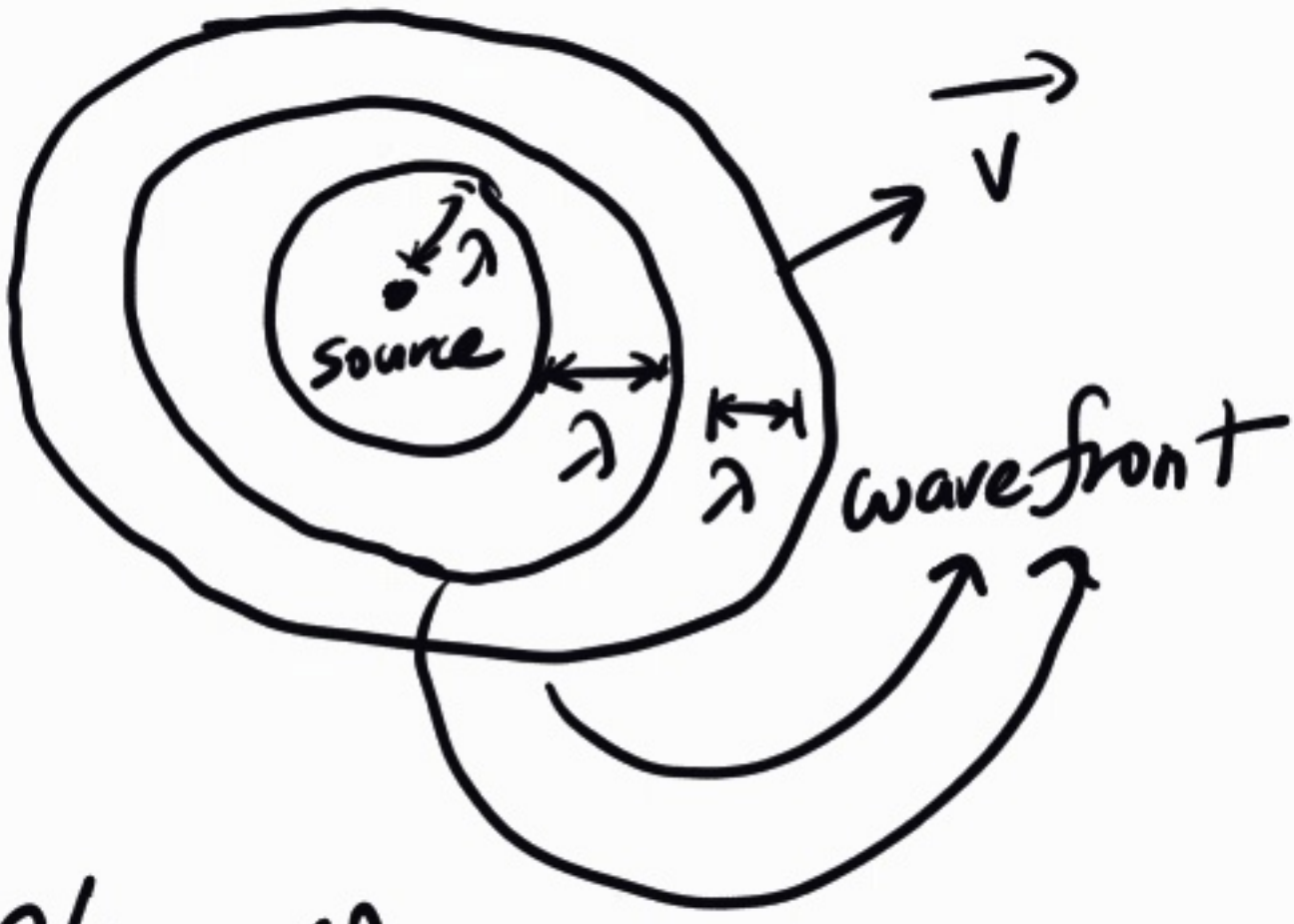
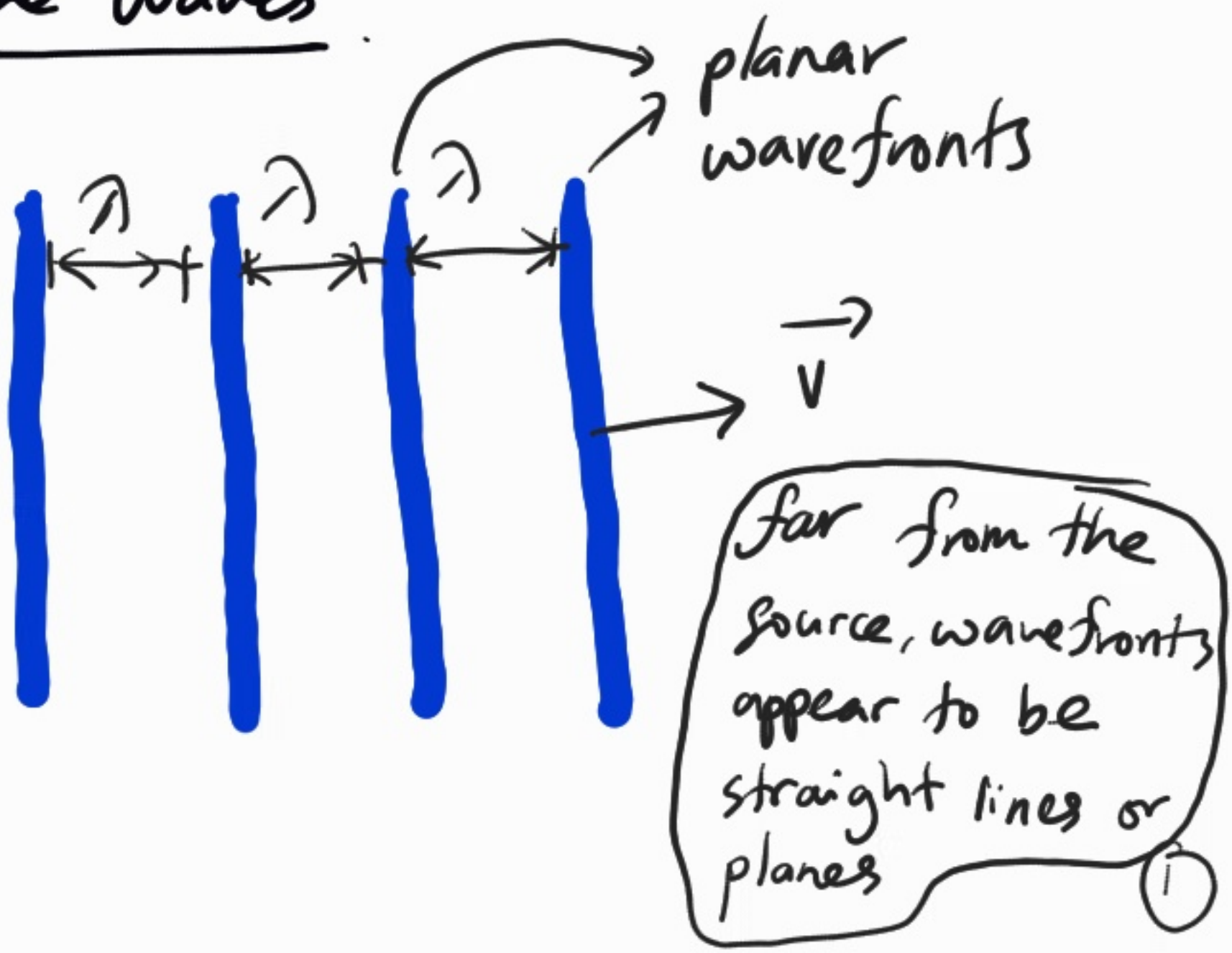
Circular, Spherical & Plane Waves



Plane Waves



Intensity : Spherical Waves

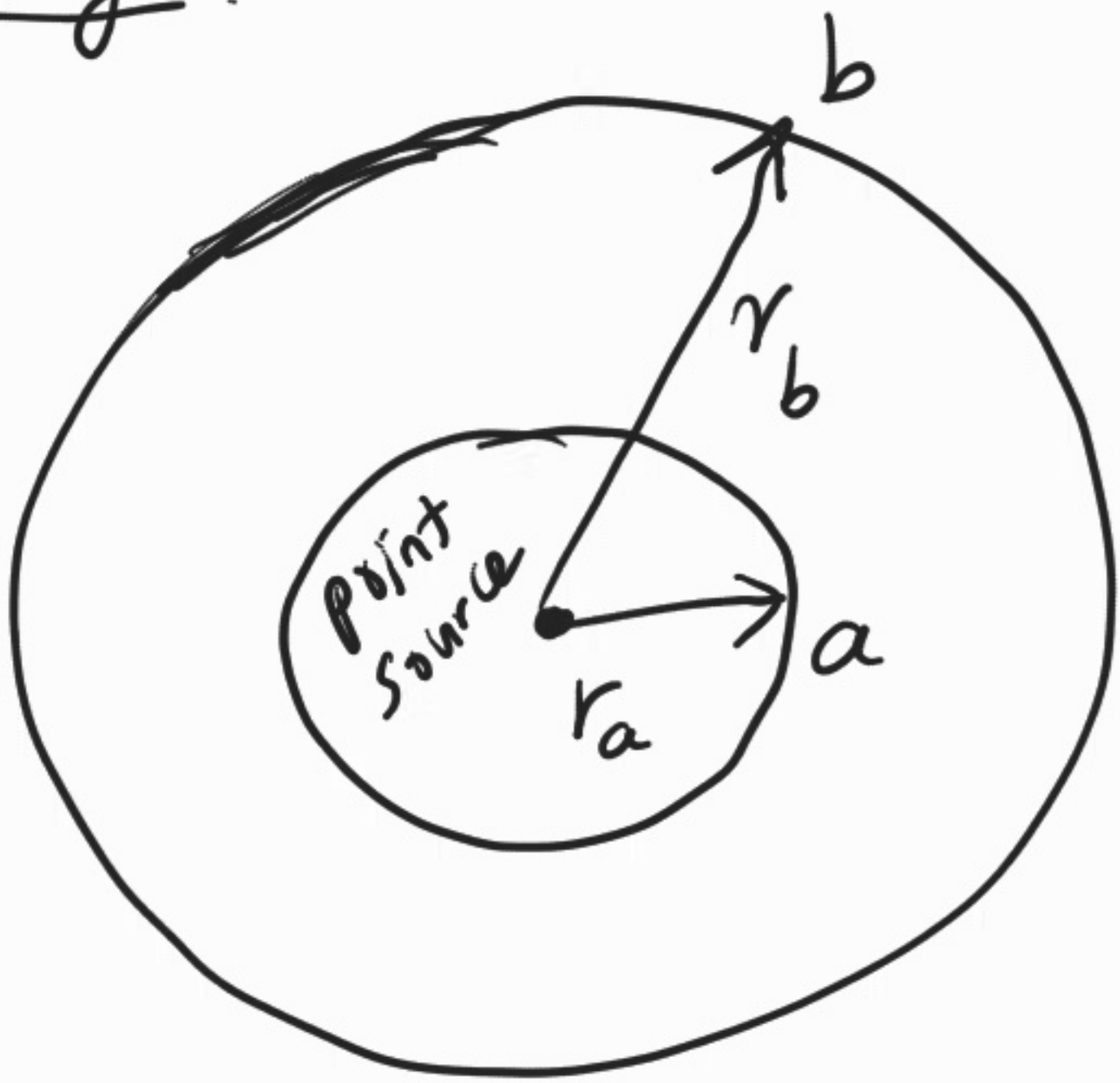
- 3-dimensional
- Point source emits sound wave of power, P equally in all directions.

$$\text{Sound Intensity} = \left[\frac{\text{Sound Wave Power}}{\text{Area of a Sphere}}\right]$$

$$I = \frac{P}{4\pi r^2} \quad [w/m^2]$$



Inverse Square Law for Sound Intensity



$$\frac{I_a}{I_b} = \frac{r_b^2}{r_a^2}$$

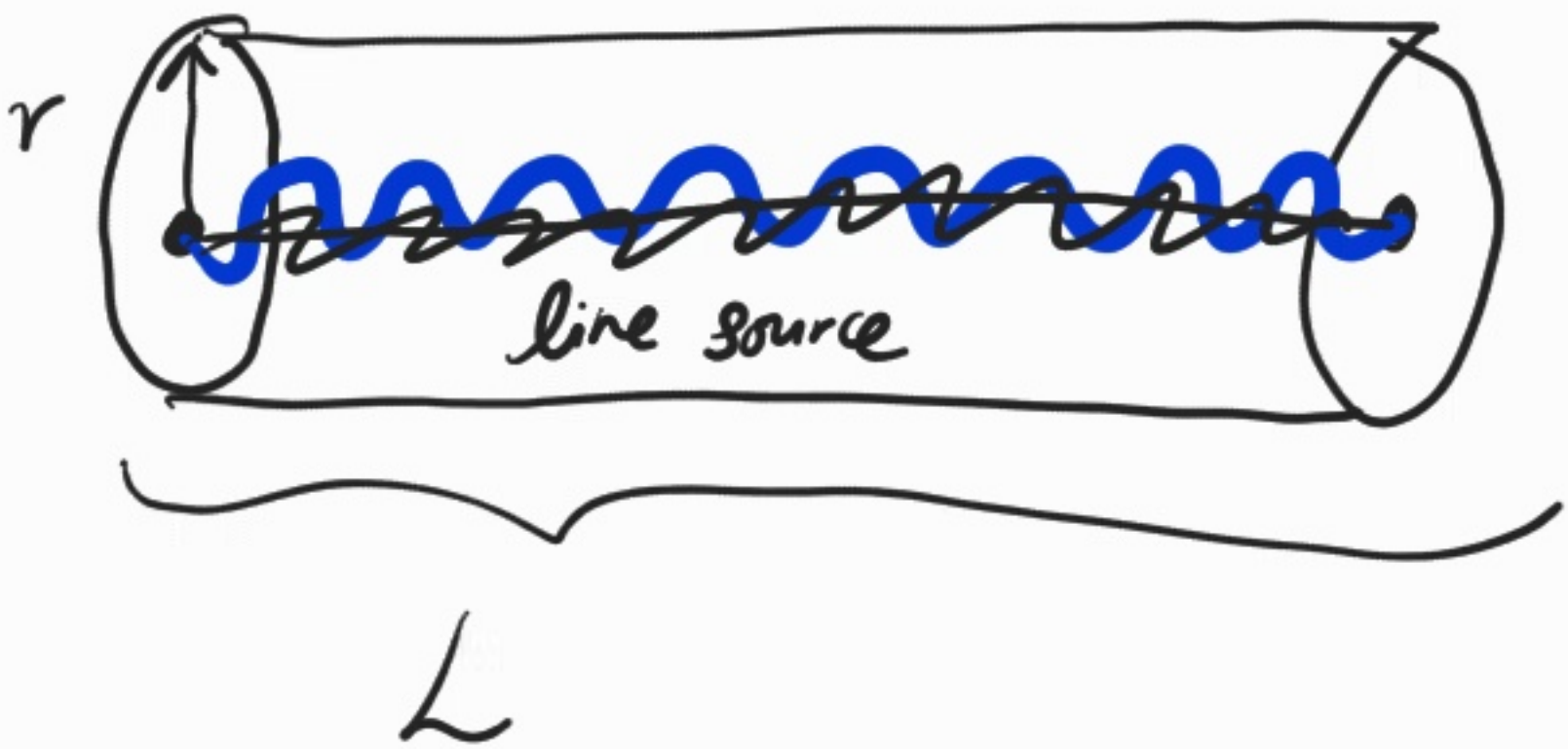
I_a → Intensity at point a

I_b → Intensity at point b

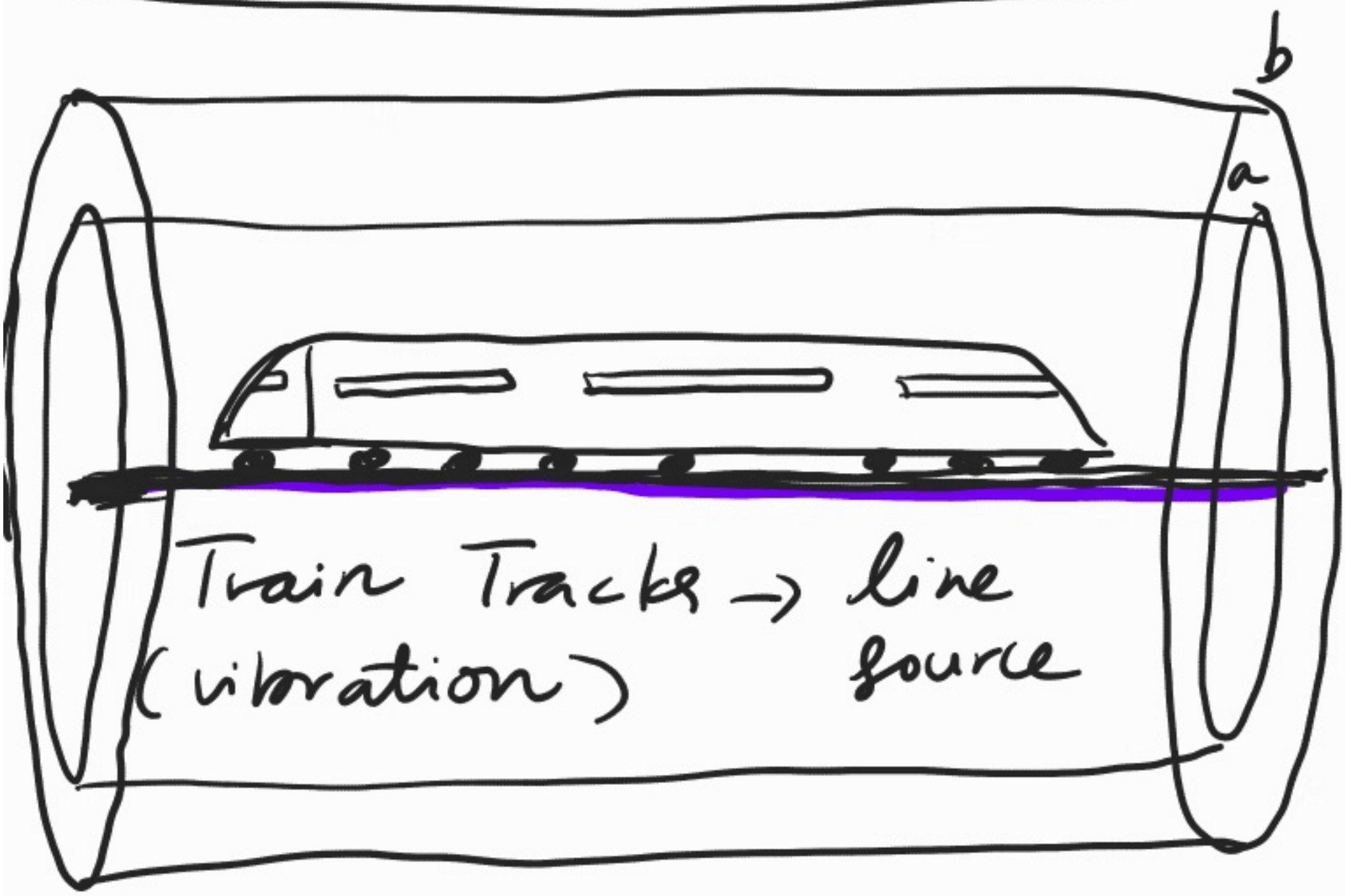
Intensity : Cylindrical Wave (2D)

- A point source emits sound wave of power P equally in all directions.

$$\text{Sound Intensity } [W/m^2] = \left[\frac{\text{Sound wave Power}}{\text{Area of lateral Surface}}\right]$$

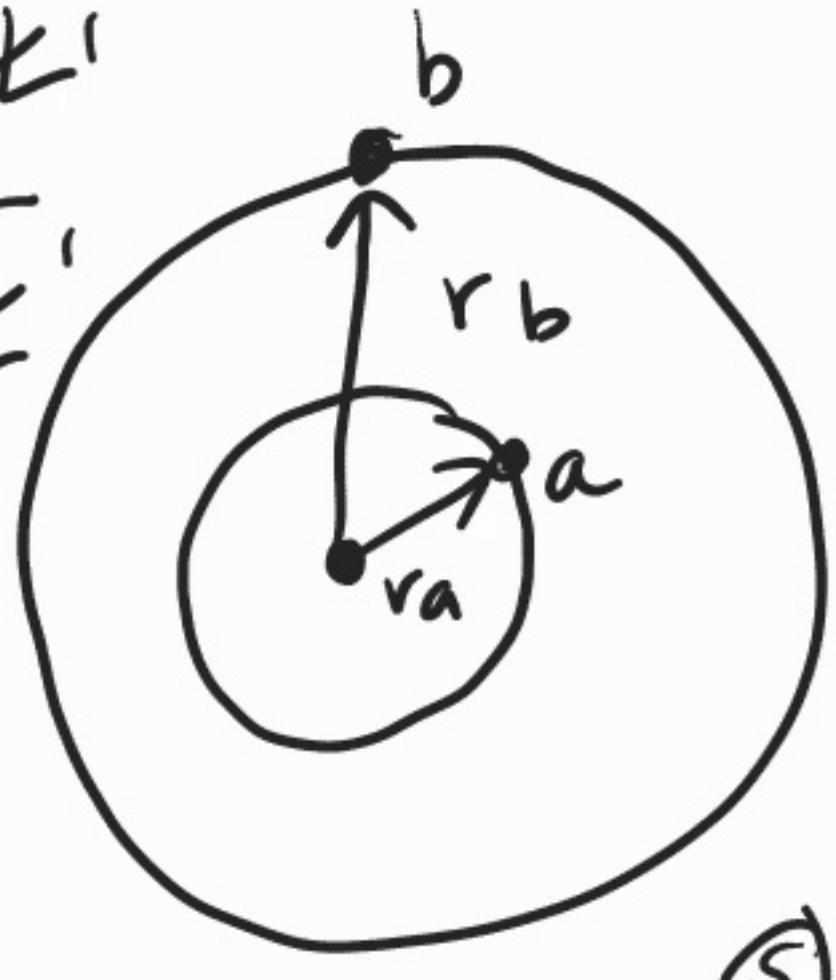


Example of a Line Source



$$\frac{I_a}{I_b} = \frac{r_b \cancel{2\pi} L'}{r_a \cancel{2\pi} L'}$$

$$= \frac{r_b}{r_a}$$



Acoustic Attenuation

- As sound propagates, $I \downarrow$ due to the spreading of E.
- measures energy loss of sound propagation in media
- there is always thermal consumption of energy caused by viscousity.

$$I(r_2) = I(r_1)\, e^{-\alpha \Delta r} \quad : \Delta r = |r_2 - r_1|$$

$\alpha \Rightarrow$ attenuation coefficient.

$I(r_1) \Rightarrow$ intensity at r_1 from the source

$I(r_2) \Rightarrow$ intensity at r_2 from the source

- Combining attenuation & spreading : $I(r_2) = I_2$ & $I(r_1) = I_1$

$$\frac{I_2}{I_1} = \left[\frac{r_1}{r_2}\right]^{n-1} e^{-\alpha \Delta r}$$

$n = 2$ for cylindrical wave

$n = 3$ for spherical wave

i/ Spreading

$$\frac{I_2}{I_1} = \left(\frac{r_1}{r_2}\right)^{n-1} \quad \text{——— (1)}$$

ii/ Attenuation

$$\frac{I_2}{I_1} = e^{-\alpha \Delta r} \quad \text{——— (2)}$$

Sound Level

- **Hearing Threshold**:
  - lowest intensity of sound that can be heard.

$$I_0 = 1 \times 10^{-12}\ W/m^2$$

- **Loudness Scale** → sound intensity is expressed in dB
  - (dimensionless)

$$\beta = (10\ dB) \log_{10}\left(\frac{I}{I_0}\right)$$

	β	I
Busy Street Traffic	70 dB	$10^{-5}\ W/m^2$

- Sound Intensity could also be found with respect to sound level

Since $\log_{10}(x) = y \rightarrow 10^{y} = x$

$\therefore I = I_0 \, 10^{\beta/10\,dB}$

Sound Level

- change in sound intensity level incorporating attenuation and spreading of sound.
- Intensity level at 2 different points :

$$\beta_1 = 10\,dB \log_{10}\left(\frac{I_1}{I_0}\right)$$

$$\beta_2 = 10\,dB \log_{10}\left(\frac{I_2}{I_0}\right)$$

$$\Delta\beta = \beta_2 - \beta_1 = 10\,dB \log_{10}\left(\frac{I_2}{I_1}\right)$$

$$= 10\,dB \log_{10}\left[\left(\frac{r_1}{r_2}\right)^{n-1} e^{-\alpha \Delta r}\right]$$

- Difference between the Intensity levels at two different points.

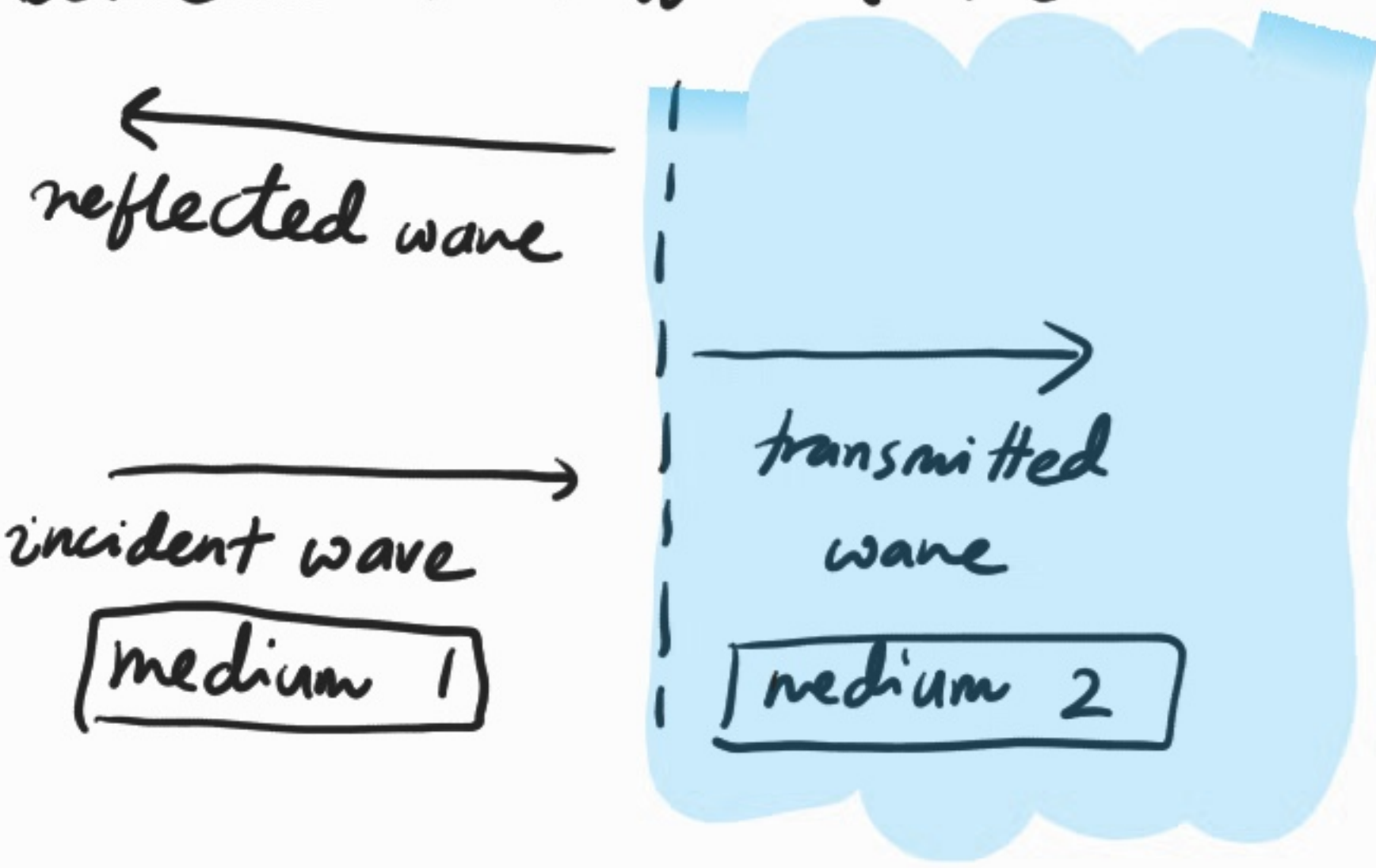
$$\Delta\beta = (10\,dB)\log_{10}\left[\left(\frac{r_2}{r_1}\right)^{n-1} e^{-\alpha\Delta r}\right]$$

$$= \left[(n-1)(10\,dB)\log_{10}\left(\frac{r_1}{r_2}\right)\right] - 4.343\,\alpha\Delta r$$

Note: $\boxed{\log_{10} e = 0.4343}$

Impedance, Reflection & Transmission

- Consider a wave being reflected & transmitted at the boundary between 2 different media.



- Impedance (Z) is a property that describes how a wave is transmitted or reflected.
- Z relies on the elastic & inertial properties of the medium.

Impedance (Z)

$$Z = \sqrt{\text{inertial property} \times \text{elastic property}}$$

- **Wave on a String:**

$$Z = \sqrt{\mu T} \quad \left(\text{linear density} \times \text{tension}\right)$$

- **Fluids:**

$$Z = \sqrt{\rho B} \quad (\text{density} \times \text{Bulk modulus})$$

- **Solid rod:**

$$Z = \sqrt{\rho Y} \quad (\text{density} \times \text{Young's Modulus})$$

Impedance (Z)

Example : μ_1, τ_1 | μ_2, τ_2

of 2 strings attached

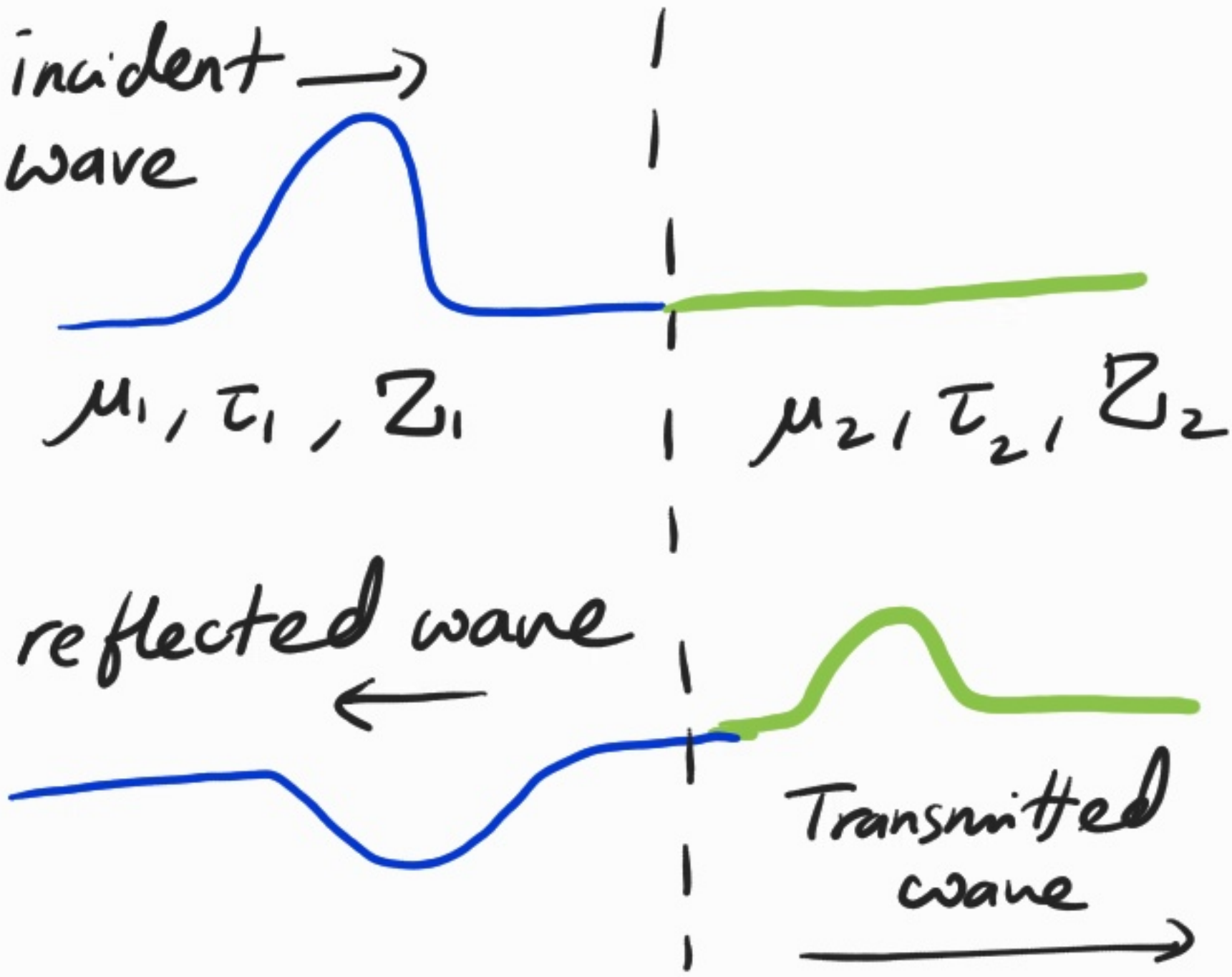
$Z_1 = \sqrt{\mu_1 \tau_1}$ $Z_2 = \sqrt{\mu_2 \tau_2}$

- **Boundary Conditions** :

→ applied to the displacement of the wave function at the boundary between the 2 media to find the relationship between

y_m of the → a) incident wave
→ b) reflected wave
↳ c) transmitted wave

Boundary Conditions



$y_{m,\text{reflected}} \leftrightarrow \underset{m}{y},\ \text{incident}$

$\underset{m}{y},\ \text{transmitted} \leftrightarrow \underset{m}{y},\ \text{incident}$